```
  GNU nano 8.7.1       producer_consumer_sem.c *
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>
#include <time.h>

#define BUFFER_SIZE 5
#define ITEMS_TO_PRODUCE 20

int buffer[BUFFER_SIZE];
int in = 0;
int out = 0;

// Semaphores and Mutex
sem_t empty_slots; // Counts empty slots in buffer
sem_t full_slots;  // Counts filled slots in buffer
pthread_mutex_t mutex; // Protects critical section / buffer a>

void *producer(void *arg) {
    for (int i = 0; i < ITEMS_TO_PRODUCE; i++) {
        int item = rand() % 100; // Produce an item

        sem_wait(&empty_slots);  // Wait for an empty slot
        pthread_mutex_lock(&mutex); // Enter critical section

        // Insert item into buffer
        buffer[in] = item;
        printf("Produced : %2d at buffer index %d\n", item, in>
        in = (in + 1) % BUFFER_SIZE;

        pthread_mutex_unlock(&mutex); // Exit critical section
        sem_post(&full_slots);  // Increment count of full slo>
```

```
 GNU nano 8.7.1          producer_consumer_sem.c *
        usleep(100000); // Sleep briefly to simulate productio>
    }
    return NULL;
}

void *consumer(void *arg) {
    for (int i = 0; i < ITEMS_TO_PRODUCE; i++) {
        sem_wait(&full_slots);  // Wait for a filled slot
        pthread_mutex_lock(&mutex); // Enter critical section

        // Remove item from buffer
        int item = buffer[out];
        printf("Consumed : %2d from buffer index %d\n", item, >
        out = (out + 1) % BUFFER_SIZE;

        pthread_mutex_unlock(&mutex); // Exit critical section
        sem_post(&empty_slots); // Increment count of empty sl>

        usleep(150000); // Sleep briefly to simulate consumpti>
    }
    return NULL;
}

int main() {
    pthread_t prod_thread, cons_thread;
    srand(time(NULL));

    // Initialize semaphores and mutex
    sem_init(&empty_slots, 0, BUFFER_SIZE);
    sem_init(&full_slots, 0, 0);
    pthread_mutex_init(&mutex, NULL);

    printf("=== Starting Producer-Consumer (Buffer Size = %d) >
```

```c
    // Create Producer and Consumer threads
    pthread_create(&prod_thread, NULL, producer, NULL);
    pthread_create(&cons_thread, NULL, consumer, NULL);

    // Wait for threads to finish
    pthread_join(prod_thread, NULL);
    pthread_join(cons_thread, NULL);

    // Cleanup
    sem_destroy(&empty_slots);
    sem_destroy(&full_slots);
    pthread_mutex_destroy(&mutex);

    printf("=== Producer-Consumer Completed Successfully ===\n>
    return 0;
}
```

```
PS C:\Users\Gireesh> wsl
gireesh@Kannigantiii:/mnt/c/Users/Gireesh$ cd ~/oslab
gireesh@Kannigantiii:~/oslab$ nano producer_consumer_sem.c
```

```
gireesh@Kannigantiii:~/oslab$ gcc -Wall producer_consumer_sem.c
 -lpthread -o producer_consumer_sem
./producer_consumer_sem
=== Starting Producer-Consumer (Buffer Size = 5) ===
Produced : 62 at buffer index 0
Consumed : 62 from buffer index 0
Produced : 63 at buffer index 1
Consumed : 63 from buffer index 1
Produced :  4 at buffer index 2
Consumed :  4 from buffer index 2
Produced : 89 at buffer index 3
Produced : 94 at buffer index 4
Consumed : 89 from buffer index 3
Produced : 15 at buffer index 0
Consumed : 94 from buffer index 4
Produced : 70 at buffer index 1
Produced : 52 at buffer index 2
Consumed : 15 from buffer index 0
Produced : 95 at buffer index 3
Consumed : 70 from buffer index 1
Produced : 62 at buffer index 4
Produced : 90 at buffer index 0
Consumed : 52 from buffer index 2
Produced : 27 at buffer index 1
Consumed : 95 from buffer index 3
Produced : 38 at buffer index 2
Produced : 29 at buffer index 3
Consumed : 62 from buffer index 4
Produced : 60 at buffer index 4
Consumed : 90 from buffer index 0
Produced : 46 at buffer index 0
Consumed : 27 from buffer index 1
Produced : 30 at buffer index 1
Consumed : 38 from buffer index 2
Produced : 94 at buffer index 2
Consumed : 29 from buffer index 3
Produced :  0 at buffer index 3
```

```
Consumed : 46 from buffer index 0
Consumed : 30 from buffer index 1
Consumed : 94 from buffer index 2
Consumed :  0 from buffer index 3
Consumed : 74 from buffer index 4
=== Producer-Consumer Completed Successfully ===
gireesh@Kannigantiii:~/oslab$
```